The Future of Artificial Intelligence in Zimbabwe

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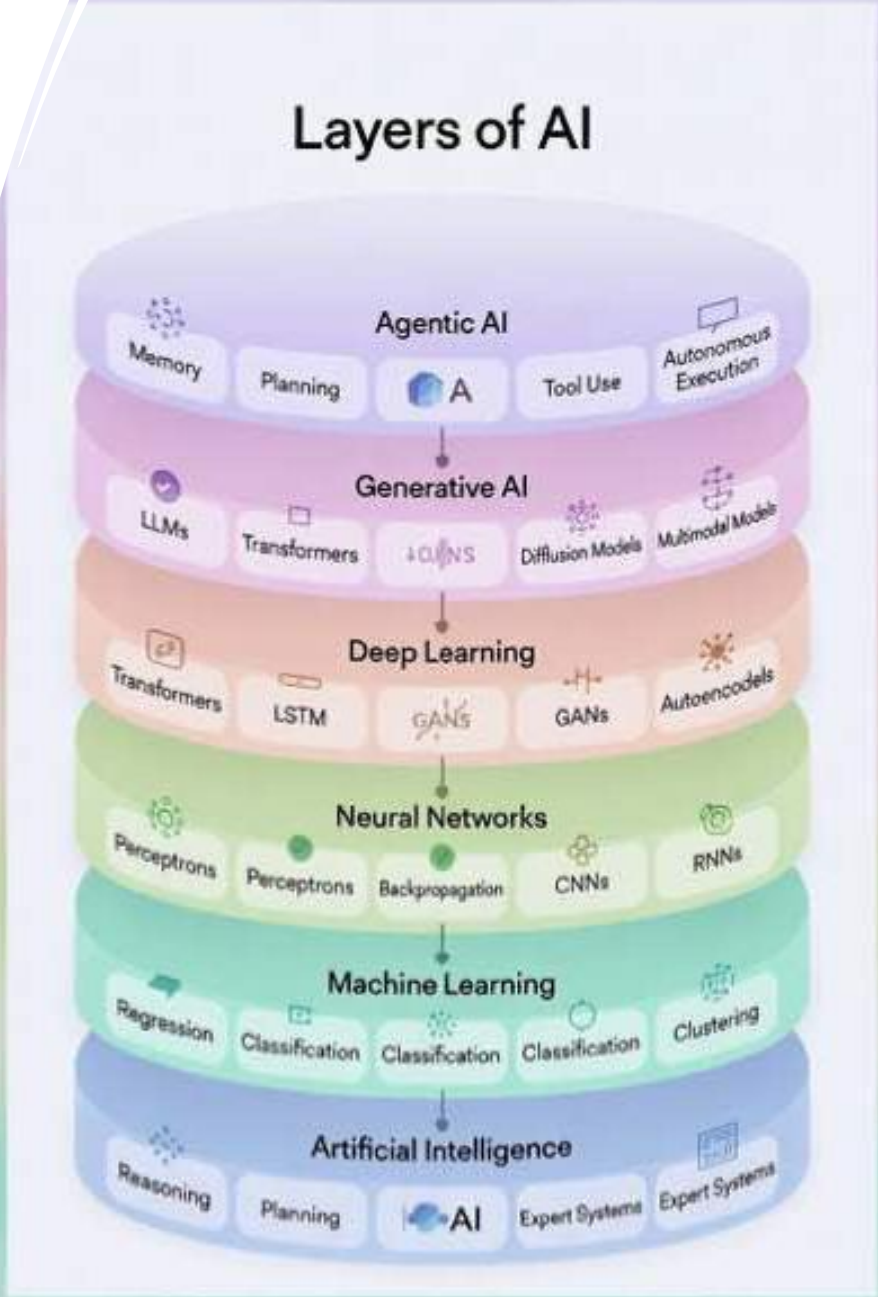
Introduction

- Definition of Artificial Intelligence
- Global importance of AI
- Why AI matters for Zimbabwe



What is Artificial Intelligence?

Machines performing tasks that require human intelligence

Learning from data

Automation and decision making

Global AI Trends

Rapid growth in AI adoption

AI transforming economies

Digital transformation worldwide

AI in Africa

Growing AI ecosystems

Emerging research hubs

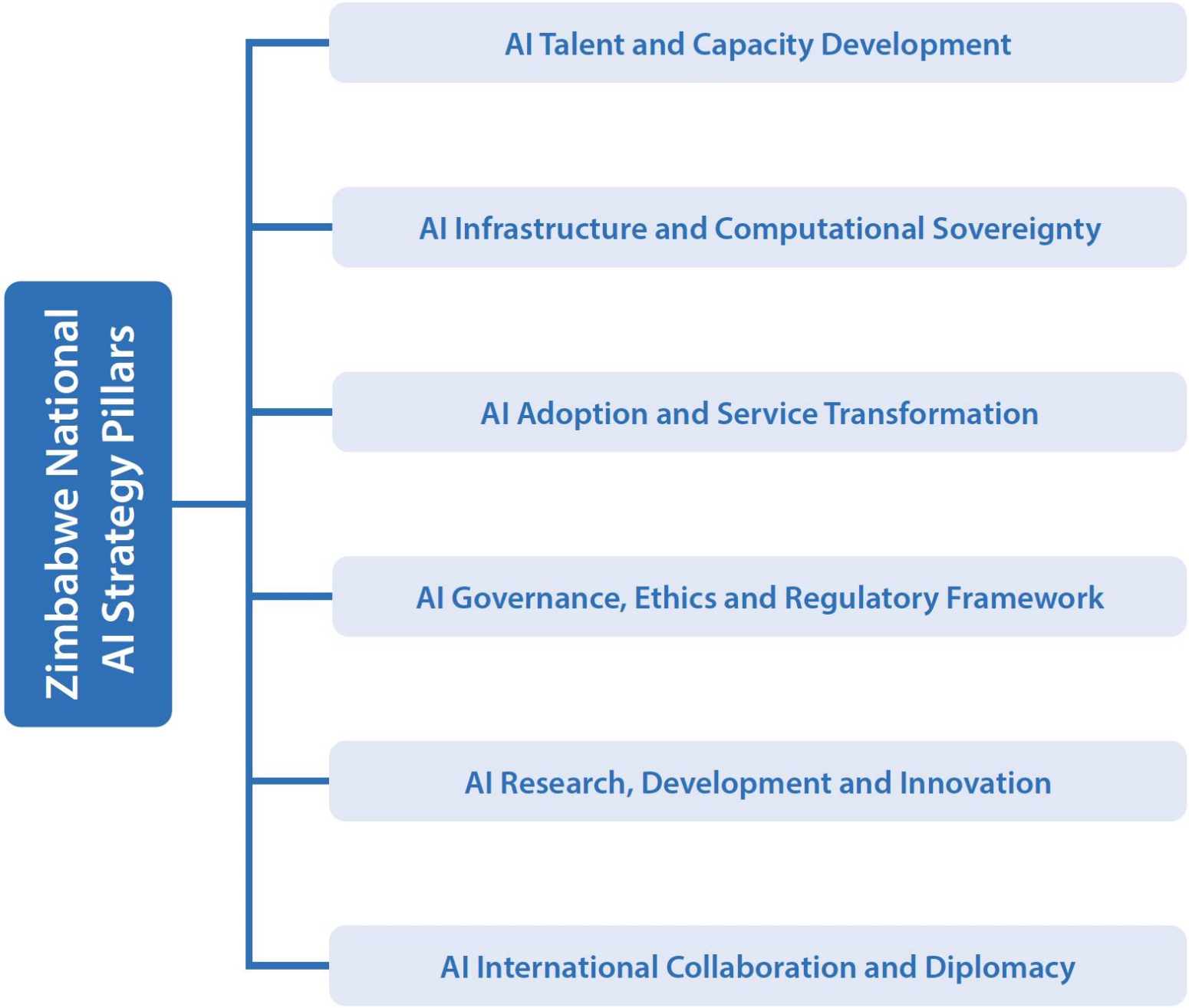
Government AI strategies

AI in Zimbabwe Today

National Artificial Intelligence Strategy

Innovation hubs at universities

Growing ICT sector

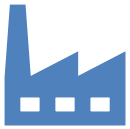

Zimbabwe National AI Strategy Pillars
AI Talent and Capacity Development
AI Infrastructure and Computational Sovereignty
AI Adoption and Service Transformation
AI Governance, Ethics and Regulatory Framework
AI Research, Development and Innovation
AI International Collaboration and Diplomacy

AI in Different economies

Agriculture

Finance

Education



Manufacturing

- Policy
- Environment
- Mining
- Materials

AI in Healthcare

- Medical diagnostics
- Disease detection
- Telemedicine

DIAGNOSTICS
TELEMEDICINE
AI IN HEALTHCARE
Transforming Healthcare in Africa
MENTAL HEALTH
PREDICTIVE ANALYTICS

Hospital Set Up in Zimbabwe

Quinary

Central

Provinces

Districts

Clinics

Opportunities for Zimbabwe

- Economic growth
- Job creation
- Technology innovation

Efforts by Zimbabwe for Health in Future

- **1. Digital Health & Information Systems**
- **2. Infrastructure & Utility Systems**
- **3. Management & Operational Systems**
- **4. Regulatory & Referral Framework**
- **5. Key Modernization Projects (2025)**

1. Digital Health & Information Systems

- **Impilo EMR System:** The Ministry of Health and Child Care (MoHCC) is rolling out this comprehensive, locally-adapted electronic health record system to standardize patient-centric care across public health facilities.
- **Ksatria HIS:** A modular Hospital Information System used to move hospitals from paper-based, fragmented workflows to structured digital operations (records, billing, pharmacy).
- **OmniSol HIS:** A widely used system (in over 900 sites) for managing electronic medical records, ward management, inventory, and billing.
- **DHIS2 Integration:** Systems are designed to align with national reporting standards (DHIS2) to provide accurate, real-time data on disease prevalence and resource availability.

2. Infrastructure & Utility Systems

- **Solar-for-Health (Solarization):** To ensure power continuity, hospitals (e.g., Gwanda Provincial Hospital) are installing large-scale solar photovoltaic panels, inverters, and battery storage systems, prioritizing maternity and pharmacy blocks.
- **Water Security:** Modernization projects include installing water purification plants and drilling boreholes to ensure continuous water supply, crucial for infection control.

3. Management & Operational Systems

- **5S-Kaizen-TQM:** JICA has introduced this Japanese management approach to 14+ hospitals to improve workspace organization, workflow efficiency, and service quality.
- **Performance-Based Financing (HDF-RBF):** An innovative funding model used in rural health centres to purchase supplies and improve quality, where funding is linked to the number of clients served.

4. Regulatory & Referral Framework

- **Registration & Licensing:** Hospitals must register departments (e.g., X-ray, laboratory) with the Health Professions Authority (HPA).
- **Pyramidal Referral System:** The health system operates on a four-tier structure: Primary (rural clinics), Secondary (district hospitals), Tertiary (provincial hospitals), and Quaternary (central/teaching hospitals).

5. Key Modernization Projects (2025)

- **Presidential Hospital Refurbishment Program:** An initiative focused on upgrading public referral centers (e.g., Parirenyatwa Group of Hospitals) to modern, high-tech standards.
- **Belarus Cooperation Deal:** A bilateral partnership aimed at supplying advanced medical equipment and enhancing local pharmaceutical production to reduce import dependency.

Challenges

- Limited infrastructure
- Skills shortage
- Research funding

AI Skills Development

- University AI programs
- STEM education
- Technical training

Research and Innovation

- AI research centres
- University–industry collaboration

Policy and Governance

- AI regulation
- Ethical AI
- Data governance

Vision for Zimbabwe

- Regional AI hub
- Digital economy
- Innovation leadership

Conclusion

- AI is a transformative opportunity
- Requires collaboration
- Investment in skills and research